

solution.

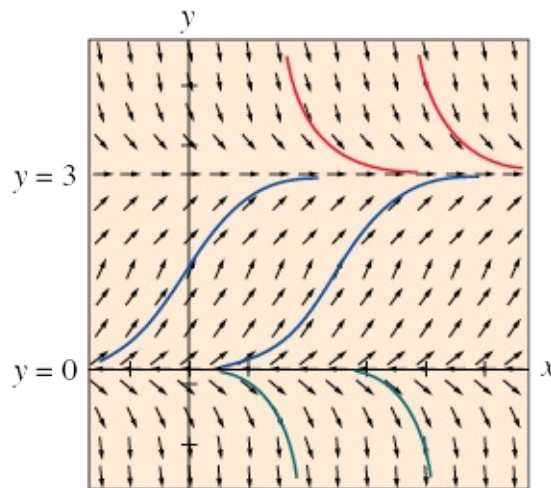


FIGURE 2.1.11 Translated solution curves of an autonomous DE

Hence, if $y(x)$ is a solution of the initial-value problem $dy/dx = f(y)$, $y(0) = y_0$, then $y_1(x) = y(x - x_0)$ is a solution of the IVP $dy/dx = f(y)$, $y(x_0) = y_0$. For example, it is easy to verify that $y(x) = e^x$, $-\infty < x < \infty$, is a solution of the IVP $dy/dx = y$, $y(0) = 1$ and so a solution $y_1(x)$ of, say, $dy/dx = y$, $y(4) = 1$ is $y(x) = e^x$ translated 4 units to the right:

$$y_1(x) = y(x - 4) = e^{x-4}, \quad -\infty < x < \infty.$$

2.1 Exercises

Answers to selected odd-numbered problems begin on page ANS-2.

2.1.1 Direction Fields

In Problems 1–4, reproduce the given computer-generated direction field. Then sketch, by hand, an approximate solution curve that passes through each of the indicated points. Use different colored pencils for each solution curve.

1. $\frac{dy}{dx} = x^2 - y^2$

(a) $y(-2) = 1$

(b) $y(3) = 0$

(c) $y(0) = 2$

(d) $y(0) = 0$

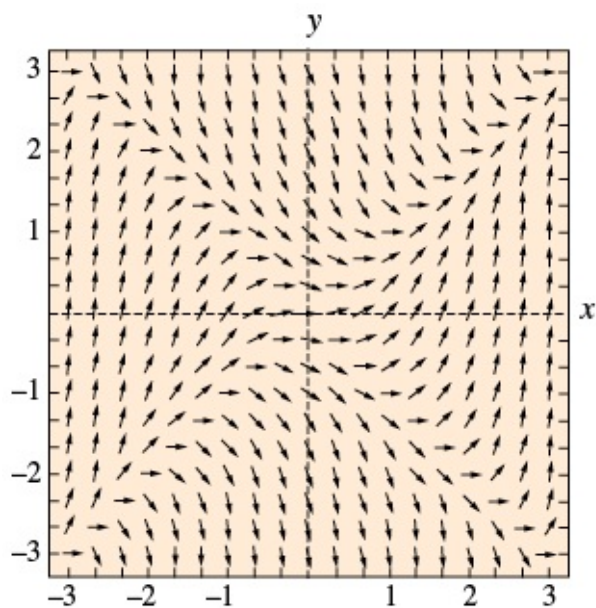


FIGURE 2.1.12 Direction field for Problem 1

2. $\frac{dy}{dx} = e^{-0.01xy^2}$

(a) $y(-6) = 0$

(b) $y(0) = 1$

(c) $y(0) = -4$

(d) $y(8) = -4$

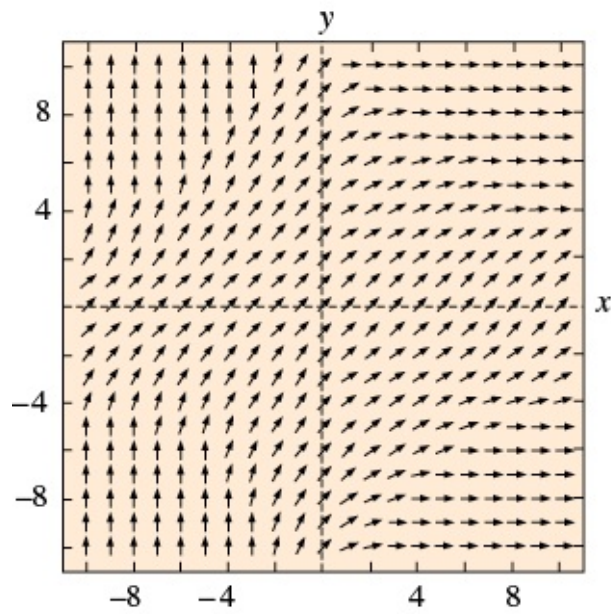


FIGURE 2.1.13 Direction field for Problem 2

3. $\frac{dy}{dx} = 1 - xy$

(a) $y(0) = 0$

(b) $y(-1) = 0$

(c) $y(2) = 2$

(d) $y(0) = -4$

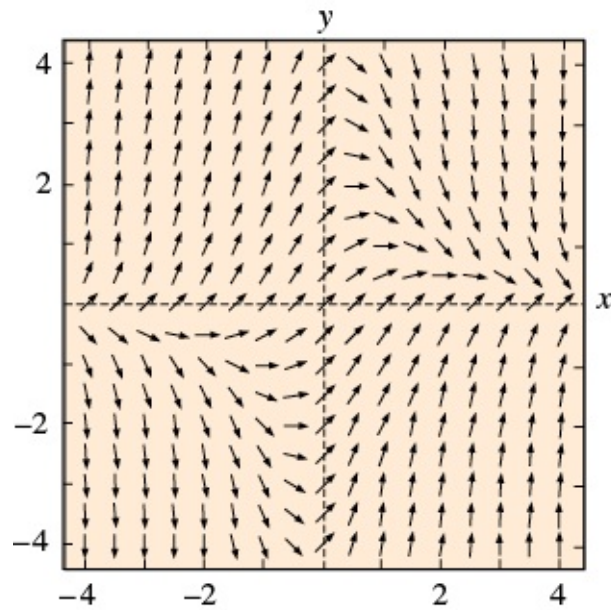


FIGURE 2.1.14 Direction field for Problem 3

4. $\frac{dy}{dx} = (\sin x) \cos y$

(a) $y(0) = 1$

(b) $y(1) = 0$

(c) $y(3) = 3$

(d) $y(0) = -\frac{5}{2}$

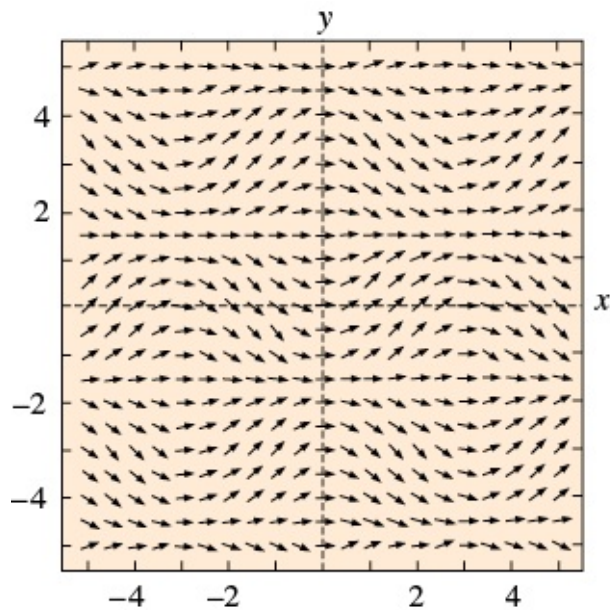


FIGURE 2.1.15 Direction field for Problem 4

In Problems 5–12, use computer software to obtain a direction field for the given differential equation. By hand, sketch an approximate solution curve passing through each of the given points.

5. $y' = x$

(a) $y(0) = 0$

(b) $y(0) = -3$

6. $y' = x + y$

(a) $y(-2) = 2$

(b) $y(1) = -3$

7. $y \frac{dy}{dx} = -x$

(a) $y(1) = 1$

(b) $y(0) = 4$

8. $\frac{dy}{dx} = \frac{1}{y}$

(a) $y(0) = 1$

(b) $y(-2) = -1$

9. $\frac{dy}{dx} = 0.2x^2 + y$

(a) $y(0) = \frac{1}{2}$

(b) $y(2) = -1$

10. $\frac{dy}{dx} = xe^y$

(a) $y(0) = -2$

(b) $y(1) = 2.5$

11. $y' = y - \cos \frac{\pi}{2}x$

(a) $y(2) = 2$

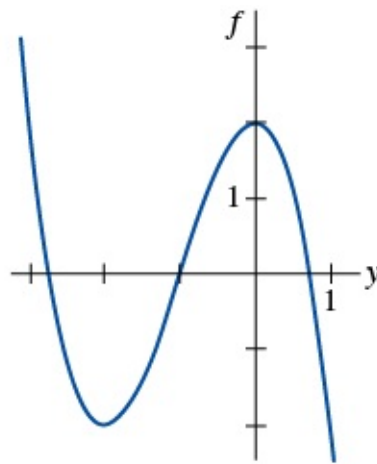
(b) $y(-1) = 0$

12. $\frac{dy}{dx} = 1 - \frac{y}{x}$

(a) $y(-\frac{1}{2}) = 2$

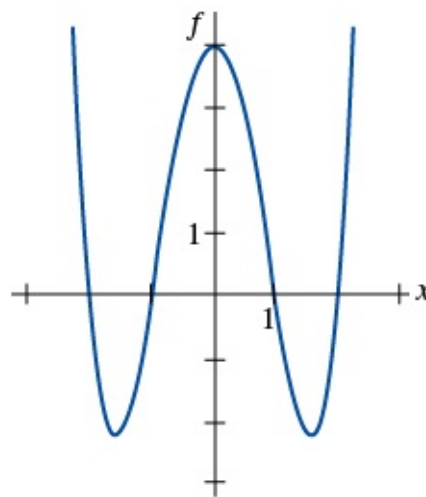
(b) $y(\frac{3}{2}) = 0$

In Problems 13 and 14, the given figures represent the graph of $f(y)$ and $f(x)$, respectively. By hand, sketch a direction field over an appropriate grid for $dy/dx = f(y)$ (Problem 13) and then for $dy/dx = f(x)$ (Problem 14).



13.

FIGURE 2.1.16 Graph for Problem 13



14.

FIGURE 2.1.17 Graph for Problem 14

15. In parts (a) and (b) sketch **isoclines** $f(x, y) = c$ (see the *Remarks* on page 36) for the given differential equation using the indicated values of c . Construct a direction field over a grid by carefully drawing lineal elements with the appropriate slope at chosen points on each isocline. In each case, use this rough direction field to sketch an approximate solution curve for the IVP consisting of the DE and the initial condition $y(0) = 1$.

(a) $dy/dx = x + y$; c an integer satisfying $-5 \leq c \leq 5$

(b) $dy/dx = x^2 + y^2$; $c = \frac{1}{4}$, $c = 1$, $c = \frac{9}{4}$, $c = 4$

Discussion Problems

16.

(a) Consider the direction field of the differential equation $dy/dx = x(y - 4)^2 - 2$, but do not use technology to obtain it. Describe the slopes of the lineal elements on the lines $x = 0$, $y = 3$, $y = 4$, and $y = 5$.

(b) Consider the IVP $dy/dx = x(y - 4)^2 - 2$, $y(0) = y_0$, where $y_0 < 4$. Can a solution $y(x) \rightarrow \infty$ as $x \rightarrow \infty$? Based on the information in part (a), discuss.

17. For a first-order DE $dy/dx = f(x, y)$, a curve in the plane defined by $f(x, y) = 0$ is called a **nullcline** of the equation, since a lineal element at a point on the curve has zero slope. Use computer software to obtain a direction field over a rectangular grid of points for $dy/dx = x^2 - 2y$, and then superimpose the graph of the nullcline $y = \frac{1}{2}x^2$ over the direction field. Discuss the behavior of solution curves in regions of the plane defined by $y < \frac{1}{2}x^2$ and by $y > \frac{1}{2}x^2$. Sketch some approximate solution curves. Try to generalize your observations.

18.

(a) Identify the nullclines (see Problem 17) in Problems 1, 3, and 4. With a colored pencil, circle any lineal elements in **FIGURES 2.1.12, 2.1.14, and 2.1.15** that you think may be a lineal element at a point on a nullcline.

(b) What are the nullclines of an autonomous first-order DE?

2.1.2 Autonomous First-Order DEs

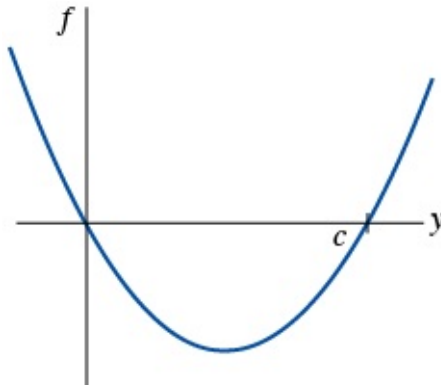
19. Consider the autonomous first-order differential equation $dy/dx = y - y^3$ and the initial condition $y(0) = y_0$. By hand, sketch the graph of a typical solution $y(x)$ when y_0 has the given values.
- (a) $y_0 > 1$
 - (b) $0 < y_0 < 1$
 - (c) $-1 < y_0 < 0$
 - (d) $y_0 < -1$
20. Consider the autonomous first-order differential equation $dy/dx = y^2 - y^4$ and the initial condition $y(0) = y_0$. By hand, sketch the graph of a typical solution $y(x)$ when y_0 has the given values.
- (a) $y_0 > 1$
 - (b) $0 < y_0 < 1$
 - (c) $-1 < y_0 < 0$
 - (d) $y_0 < -1$

In Problems 21–28, find the critical points and phase portrait of the given autonomous first-order differential equation. Classify each critical point as asymptotically stable, unstable, or semi-stable. By hand, sketch typical solution curves in the regions in the xy -plane determined by the graphs of the equilibrium solutions.

21. $\frac{dy}{dx} = y^2 - 3y$
22. $\frac{dy}{dx} = y^2 - y^3$

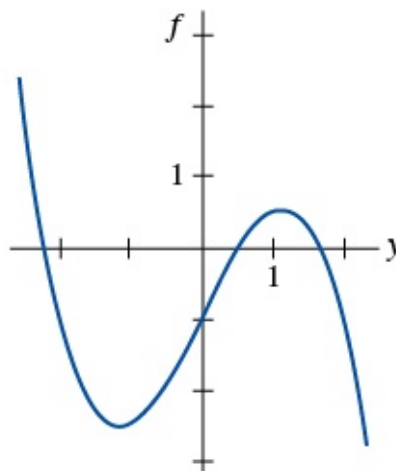
23. $\frac{dy}{dx} = (y - 2)^4$
24. $\frac{dy}{dx} = 10 + 3y - y^2$
25. $\frac{dy}{dx} = y^2(4 - y^2)$
26. $\frac{dy}{dx} = y(2 - y)(4 - y)$
27. $\frac{dy}{dx} = y \ln(y + 2)$
28. $\frac{dy}{dx} = \frac{ye^y - 9y}{e^y}$

In Problems 29 and 30, consider the autonomous differential equation $dy/dx = f(y)$, where the graph of f is given. Use the graph to locate the critical points of each differential equation. Sketch a phase portrait of each differential equation. By hand, sketch typical solution curves in the subregions in the xy -plane determined by the graphs of the equilibrium solutions.



29.

FIGURE 2.1.18 Graph for Problem 29



30.

FIGURE 2.1.19 Graph for Problem 30

Discussion Problems

31. Consider the autonomous DE $dy/dx = (2/\pi)y - \sin y$. Determine the critical points of the equation. Discuss a way of obtaining a phase portrait of the equation. Classify the critical points as asymptotically stable, unstable, or semi-stable.
32. A critical point c of an autonomous first-order DE is said to be **isolated** if there exists some open interval that contains c but no other critical point. Discuss: Can there exist an autonomous DE of the form given in (1) for which every critical point is nonisolated? Do not think profound thoughts.
33. Suppose that $y(x)$ is a nonconstant solution of the autonomous equation $dy/dx = f(y)$ and that c is a critical point of the DE. Discuss: Why can't the graph of $y(x)$ cross the graph of the equilibrium solution $y = c$? Why can't $f(y)$ change signs in one of the subregions discussed on page 38? Why can't $y(x)$ be oscillatory or have a relative extremum (maximum or minimum)?
34. Suppose that $y(x)$ is a solution of the autonomous equation $dy/dx = f(y)$ and is bounded above and below by two

consecutive critical points $c_1 < c_2$, as in subregion R_2 of Figure 2.1.5(b). If $f(y) > 0$ in the region, then $\lim_{x \rightarrow \infty} y(x) = c_2$. Discuss why there cannot exist a number $L < c_2$ such that $\lim_{x \rightarrow \infty} y(x) = L$. As part of your discussion, consider what happens to $y'(x)$ as $x \rightarrow \infty$.

35. Using the autonomous equation (1), discuss how it is possible to obtain information about the location of points of inflection of a solution curve.
36. Consider the autonomous DE $dy/dx = y^2 - y - 6$. Use your ideas from Problem 35 to find intervals on the y -axis for which solution curves are concave up and intervals for which solution curves are concave down. Discuss why *each* solution curve of an initial-value problem of the form $dy/dx = y^2 - y - 6$, $y(0) = y_0$, where $-2 < y_0 < 3$, has a point of inflection with the same y -coordinate. What is that y -coordinate? Carefully sketch the solution curve for which $y(0) = -1$. Repeat for $y(0) = 2$.
37. Suppose the autonomous DE in (1) has no critical points. Discuss the behavior of the solutions.

Mathematical Models

38. **Population Model** The differential equation in Example 3 is a well-known population model. Suppose the DE is changed to

$$\frac{dP}{dt} = P(aP - b),$$

where a and b are positive constants. Discuss what happens to the population P as time t increases.

39. **Population Model** Another population model is given by

$$\frac{dP}{dt} = kP - h,$$

where $h > 0$ and $k > 0$ are constants. For what initial values $P(0) = P_0$ does this model predict that the population will go extinct?

- 40. Terminal Velocity** The autonomous differential equation

$$m \frac{dv}{dt} = mg - kv,$$

where k is a positive constant of proportionality called the drag coefficient and g is the acceleration due to gravity, is a model for the instantaneous velocity v of a body of mass m that is falling under the influence of gravity. Because the term $-kv$ represents air resistance or drag, the velocity of a body falling from a great height does not increase without bound as time t increases. Use a phase portrait of the differential equation to find the limiting, or terminal, velocity of the body. Explain your reasoning. See page 24.

- 41. Terminal Velocity** In Problem 17 of Exercises 1.3, we indicated that for high-speed motion of a body, air resistance is taken to be proportional to a power of its instantaneous velocity v . If we take air resistance to be proportional to v^2 , then the mathematical model for the instantaneous velocity of a falling body of mass m in Problem 40 becomes

$$m \frac{dv}{dt} = mg - kv^2,$$

where $k > 0$. Use a phase portrait to find the terminal velocity of the body. Explain your reasoning. See page 27.

- 42. Chemical Reactions** When certain kinds of chemicals are combined, the rate at which a new compound is formed is

governed by the differential equation

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X),$$

where $k > 0$ is a constant of proportionality and $\beta > \alpha > 0$.

Here $X(t)$ denotes the number of grams of the new compound formed in time t . See pages 21 and 22.

(a) Use a phase portrait of the differential equation to predict the behavior of X as $t \rightarrow \infty$.

(b) Consider the case when $\alpha = \beta$. Use a phase portrait of the differential equation to predict the behavior of X as $t \rightarrow \infty$ when $X(0) < \alpha$. When $X(0) > \alpha$.

(c) Verify that an explicit solution of the DE in the case when $k = 1$ and $\alpha = \beta$ is $X(t) = \alpha - 1/(t + c)$. Find a solution satisfying $X(0) = \alpha/2$. Find a solution satisfying $X(0) = 2\alpha$. Graph these two solutions. Does the behavior of the solutions as $t \rightarrow \infty$ agree with your answers to part (b)?